

Run 03 | Business exit

Global (including India)

IV + controls

Target Entrepreneurial/business exit in the past 12 months; binary 1 vs. 0
Sample 1,331,428 respondents **Coverage** 89 countries; 2015–2022
Positive cases 61,654 (4.63%) **Case time** 40m 19s
Train / val. / test 905,374 / 159,770 / 266,284 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Fear of failure, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.7998). Highest F1: Random Forest (0.2603). For XGBoost, adding controls changed ROC-AUC by +0.0124 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.7998	0.2055	91.24	21.08	32.49	0.2557	26s
Random Forest	0.7950	0.2209	91.40	21.63	32.68	0.2603	4m 45s
CatBoost	0.7937	0.1985	90.42	19.67	34.68	0.2510	3m 28s
H2O GBM	0.7927	0.1946	90.40	19.49	34.27	0.2485	1m 50s
Lasso	0.7706	0.1472	89.18	16.18	31.97	0.2149	6m 56s
Elastic Net	0.7706	0.1472	89.18	16.18	32.00	0.2150	7m 02s
Logistic Regression	0.7704	0.1468	89.20	16.25	32.09	0.2158	1m 23s
Ridge	0.7698	0.1460	89.18	16.21	32.08	0.2154	8m 31s
Naive Bayes	0.7023	0.1009	86.85	11.79	28.39	0.1666	4m 21s
Decision Tree	0.5000	0.0463	4.63	4.63	100.00	0.0885	36s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Age	3,377.41	12.05
2	Household size	2,082.89	7.43
3	Occupational status	1,563.47	5.58
4	Entrepreneurial intention	1,502.57	5.36
5	Education attainment	1,229.65	4.39
6	Household income	1,159.89	4.14
7	Business angel	1,088.00	3.88
8	Self-efficacy	603.15	2.15

Share is the percentage of total forest impurity importance across all 64 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	547.35	1.95
2	Prosperity index score	501.03	1.79
3	Living conditions	500.74	1.79

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 5 batch, 29 September 2026; 03_obj5_global_iv_cv_EXIT_ENT_harmonized. Scores and importance are taken directly from saved CSV outputs.